

Josephine Baggen

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Education

Ph.D. Astronomy, Yale University, USA Expected 2027

Thesis: Connecting the Dots with the James Webb Space Telescope:

The Formation of Massive Compact Galaxies and Black Holes in the First Billion Years

Advisor: Prof.dr. P. van Dokkum

Master Astronomy & Cosmology, Leiden University, The Netherlands 2020–2022

Graduated with Honours

Thesis II: The highest redshift radio galaxies

Advisor: Prof.dr. H.J.A. Röttgering, Dr. A. Gloudemans

Thesis I: Transformation of Galaxy Morphology and Alignment in Galaxy Clusters

Advisor: Prof.dr. H. Hoekstra, Dr. Y.M. Bahé

Bachelor Astronomy, Leiden University, The Netherlands 2017–2020

Thesis: Hyper-velocity Stars generated by the Large Magellanic Cloud

Advisor: Prof.dr. E.M. Rossi, Dr. F. Evans

Publications

My publication record includes 5 first-author papers and 16 total papers (h -index 10, over 1000 citations), with my two most cited first-author papers receiving 158 and 94 citations respectively.

[ADS library](#) 

First Authored

1. **Baggen, J. F. W.**, van Dokkum, P., et al. “Connecting the Dots: UV-bright Companions of Little Red Dots as Lyman–Werner Sources Enabling Direct-collapse Black Hole Formation” *ApJL*, 1002, L4 (2026). [doi:10.3847/2041-8213/ae58a5](https://doi.org/10.3847/2041-8213/ae58a5)
2. **Baggen, J. F. W.**, van den Bosch, F., et al. “The Impact of Baryonic Effects on the Dynamical Masses Inferred Using Satellite Kinematics” *The Open Journal of Astrophysics*, 9, 58198, (2026). [doi:10.33232/001c.158198](https://doi.org/10.33232/001c.158198)
3. **Baggen, J. F. W.**, van Dokkum, P., et al. “(Re)solving the complex multi-scale morphology and V-shaped SED of a newly discovered strongly-lensed Little Red Dot in Abell 383” *arXiv e-prints* (2025). [doi:10.48550/arXiv.2512.03239](https://doi.org/10.48550/arXiv.2512.03239)
4. **Baggen, J. F. W.**, van Dokkum, P., et al. “The Small Sizes and High Implied Densities of “Little Red Dots” with Balmer Breaks Could Explain Their Broad Emission Lines without an Active Galactic Nucleus” *ApJL*, 977, L13 (2024). [doi:10.3847/2041-8213/ad90b8](https://doi.org/10.3847/2041-8213/ad90b8)

5. **Baggen, J. F. W.**, van Dokkum, P., et al. “Sizes and Mass Profiles of Candidate Massive Galaxies Discovered by JWST at $7 < z < 9$: Evidence for Very Early Formation of the Central 100 pc of Present-day Ellipticals” *ApJL*, 955, L12 (2023). doi:10.3847/2041-8213/acf5ef

Contributing Author

1. van Dokkum, P., Brammer, G., Jennings, C., Pasha, I., **Baggen, J. F. W.** “Further Evidence for a Direct-collapse Origin of the Supermassive Black Hole at the Center of the ∞ Galaxy” *ApJL*, 990, L48 (2025). doi:10.3847/2041-8213/adfb50
2. van Dokkum, P., Brammer, G., **Baggen, J. F. W.**, et al. “The ∞ Galaxy: A Candidate Direct-collapse Supermassive Black Hole between Two Massive, Ringed Nuclei” *ApJL*, 988, L6 (2025). doi:10.3847/2041-8213/addcfe
3. Mitra, K., van den Bosch, F. C., **Baggen, J. F. W.**, Lange, J. U. “BASILISK IV. No S_8 Tension with Satellite Kinematics” *arXiv e-prints* (2025). doi:10.48550/arXiv.2512.14889
4. D’Eugenio, F., Nelson, E., Ji, X., **Baggen, J. F. W.**, et al. “Irony at $z = 6.68$: a bright AGN with forbidden Fe emission and multi-component Balmer absorption” *arXiv e-prints* (2025). doi:10.48550/arXiv.2510.00101
5. Naidu, R. P., Matthee, J., Katz, H., et al. (including **Baggen, J. F. W.**) “A ‘Black Hole Star’ Reveals the Remarkable Gas-Enshrouded Hearts of the Little Red Dots” *arXiv e-prints* (2025). doi:10.48550/arXiv.2503.16596
6. de Graaff, A., Rix, H.-W., Naidu, R. P., et al. (including **Baggen, J. F. W.**) “A remarkable Ruby: Absorption in dense gas, rather than evolved stars, drives the extreme Balmer break of a Little Red Dot at $z = 3.5$ ” *A&A*, 701, A168 (2025). doi:10.1051/0004-6361/202554681
7. Barrufet, L., Oesch, P. A., Marques-Chaves, R., Arellano-Cordova, K., **Baggen, J. F. W.**, et al. “Quiescent or dusty? Unveiling the nature of extremely red galaxies at $z > 3$ ” *MNRAS* (2025). doi:10.1093/mnras/staf013
8. Weibel, A., de Graaff, A., Setton, D. J., et al. (including **Baggen, J. F. W.**) “RUBIES Reveals a Massive Quiescent Galaxy at $z = 7.3$ ” *ApJ*, 983, 11 (2025). doi:10.3847/1538-4357/adab7a
9. Wang, B., Leja, J., de Graaff, A., et al. (including **Baggen, J. F. W.**) “RUBIES: Evolved Stellar Populations with Extended Formation Histories at $z \sim 7-8$ in Candidate Massive Galaxies Identified with JWST/NIRSpec” *ApJL*, 969, L13 (2024). doi:10.3847/2041-8213/ad55f7
10. Evans, F. A., Marchetti, T., Rossi, E. M., **Baggen, J. F. W.**, Bloot, S. “Comparing hyper-velocity star populations from the Large Magellanic Cloud and the Milky Way” *MNRAS*, 507, 4997 (2021). doi:10.1093/mnras/stab2271

Talks & Presentations

Contributed Talk, <i>Charting Cosmic Dawn</i> , Copenhagen, Denmark	2026
Lightning Talk, <i>Yale Center for the Invisible Universe Primordial Black Holes Workshop</i> , New Haven, USA	2026
Contributed Talk, <i>The First Billion Years: 5 Question in 5 Days</i> , Aspen, USA	2026
Invited Talk, <i>CSI: Sesto – Cosmology Scene Investigation in Sesto: joining efforts to solve JWST mysteries in the distant Universe</i> , Sesto, Italy	2026
Seminar Talk, <i>Swinburne University of Technology</i> , Melbourne, Australia	2025
Contributed Talk, <i>Kaba Kaba Conference: Exploring the First Billion Years of the Universe</i> , Queensland, Australia	2025
Contributed Talk, <i>The Inaugural Cosmic Frontier Center Conference - The Formation of the Earliest Cosmic Structures from a Joint Observational and Theoretical Perspective</i> , Austin, USA	2025
Seminar Talk, <i>Leiden University</i> , Leiden, The Netherlands	2025
54th Saas-Fee Talk Advanced Course, <i>Galaxies and Black Holes in the First Billion Years as seen by the JWST</i> , Saas-Fee, Switzerland	2025
Invited Talk, <i>Santa Cruz Workshop on Galaxy Formation</i> , Santa Cruz, USA	2024
Invited Discussion, <i>Santa Cruz Workshop on Galaxy Formation</i> , Santa Cruz, USA	2024
Poster, <i>European Astronomical Society Annual Meeting</i> , Padova, Italy	2024
Seminar Talk, <i>Leiden University</i> , Leiden, The Netherlands	2024
Contributed Talk, <i>RUBIES team meeting</i> , The Netherlands	2024
Poster, <i>First Light at MIT</i> , Boston, USA	2023

Press

<i>Astrobites</i> , Featured in conference report, Queensland, Australia	2025
<i>Yale Daily News</i> , Expert commentary on massive galaxy discovery	2024
<i>Event Horizon</i> , Invited guest on astronomy YouTube channel	2023

Observing Programs

James Webb Space Telescope (JWST)

GO #8047 PIs Wang, Nelson, **Baggen** and et al, “Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features” (**20.0 hours**)

GO #5545 PIs Barrufet, Arellano Cordova, **Baggen** and et al, “Dead or alive? Unveiling the nature of massive galaxies in the early Universe” (**10.5 hours**)

GO #9327 PIs Dokkum, Brammer, Pasha, and **Baggen**, “Confirmation of a direct-collapse super-massive black hole” (**2.8 hours**)

Keck Observatory

PIs Burke, Runnoe and **Baggen**, “Keck I LRISp - Coming out of their shells: constraining the geometry of Little Red Dots with spectropolarimetry ” (**1 night**)

Teaching & Mentoring

Yale University

Teaching Assistant, ASTR 180: Introduction to Relativity and Black Holes	2024
Teaching Assistant, ASTR 160: Frontiers and Controversies in Astrophysics	2023
Teaching Assistant, ASTR 130: Origins & Search for Life in the Universe	2023
Teaching Assistant, ASTR 160: Frontiers and Controversies in Astrophysics	2022

Other

Private Tutor, Physics, Mathematics, Chemistry (high school level)	2015–2021
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Service & Outreach

Referee, <i>The Astrophysical Journal</i>	2023–Present
Astronomy Student Council (ASC), Yale University	2024–Present
Astronomy on Tap, New Haven	2024
Public outreach talks for general audiences	
Leitner Family Planetarium, New Haven	2023–present
Public outreach talks for school groups and children	
Tour Guide, The Old Observatory, Leiden	2017–2021

Activities & Interests

Yale Club Tennis Team, Yale University	2023–Present
Committee Member, Student Rowing Association K.S.R.V. Njord, Leiden	2019–2021